

浙江大学实验报告

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实验名称：硬件基础实验-三色灯、光敏电阻和 BMP280 实验日期：2025.9.17

一、在 Arduino IDE 的串口监视器中读取光敏电阻的读数

```
void setup() {  
  // initialize serial communication at 115200 bits per second:  
  Serial.begin(115200);  
  //set the resolution to 12 bits (0-4095)  
  analogReadResolution(10);  
}  
void loop() {  
  // read the analog / millivolts value for pin 2:  
  int analogValue = analogRead(32);  
  int analogVolts = analogReadMilliVolts(32);  
  // print out the values you read:  
  Serial.printf("ADC analog value = %d\n", analogValue);  
  Serial.printf("ADC millivolts value = %d\n", analogVolts);  
  delay(100); // delay in between reads for clear read from serial  
}
```

将串口监视器的波特率改为 115200 后，读取电压信息。数据如图所示：

```
ADC millivolts value = 176  
ADC analog value = 10  
ADC millivolts value = 173  
ADC analog value = 11  
ADC millivolts value = 173  
ADC analog value = 10  
ADC millivolts value = 176  
ADC analog value = 11  
ADC millivolts value = 176  
ADC analog value = 11  
ADC millivolts value = 176  
ADC analog value = 11  
ADC millivolts value = 176  
ADC analog value = 10  
ADC millivolts value = 173  
ADC analog value = 11  
ADC millivolts value = 173  
ADC analog value = 11
```

二、在 Arduino IDE 的串口监视器中读取 BMP280 传感器的温度信息

```
#include <Arduino.h>
#include <Wire.h>
#include <Adafruit_BMP280.h>

#define BMP_SDA 21 //IC 引脚定义
#define BMP_SCL 22 //IC 引脚定义
Adafruit_BMP280 bmp280;

void setup() {
  Serial.begin(115200); //串口初始化
  Serial.println("Initializing BMP280"); //串口打印字符
  boolean status = bmp280.begin(0x76); //器件的 IC 地址是 0x76 或 0x77
  if (!status) {
    Serial.println("Not connected");
  }
}

void loop() {
  float temp = bmp280.readTemperature(); //定义变量读取温度值
  Serial.print("Temperature: ");
  Serial.print(temp); //打印温度值
  Serial.println(" *C");
  Serial.println("-----");
  delay(2000); //每两秒刷新一次
}
```

将串口监视器的波特率改为 115200 后，读取温度信息。

室温下的温度数据如图一所示；用手焐热传感器，温度数据如图二所示。温度明显出现抬升，说明 BMP280 传感器正常工作。

```
Temperature: 23.74 *C
-----
Temperature: 23.69 *C
-----
Temperature: 23.62 *C
-----
Temperature: 23.56 *C
-----
Temperature: 23.51 *C
-----
Temperature: 23.47 *C
-----
Temperature: 23.44 *C
-----
Temperature: 23.42 *C
-----
Temperature: 23.38 *C
-----
```

```
Temperature: 27.42 *C
-----
Temperature: 28.01 *C
-----
Temperature: 30.82 *C
-----
Temperature: 31.54 *C
-----
Temperature: 33.69 *C
-----
Temperature: 34.58 *C
-----
Temperature: 34.94 *C
-----
Temperature: 34.98 *C
-----
Temperature: 35.37 *C
-----
```

三、将三色小灯混合出不同的颜色

1. 产生黄色、紫色、青色、白色：

```
void loop() {
  analogWrite(RED, 255);
  analogWrite(GREEN, 255);
  analogWrite(BLUE, 0);
  delay(1000);
  analogWrite(RED, 255);
  analogWrite(GREEN, 0);
  analogWrite(BLUE, 255);
  delay(1000);
  analogWrite(RED, 0);
  analogWrite(GREEN, 255);
  analogWrite(BLUE, 255);
  delay(1000);
  analogWrite(RED, 255);
  analogWrite(GREEN, 255);
  analogWrite(BLUE, 255);
  delay(1000);
  analogWrite(RED, 0);
  analogWrite(GREEN, 0);
  analogWrite(BLUE, 0);
  delay(1000);
}
```

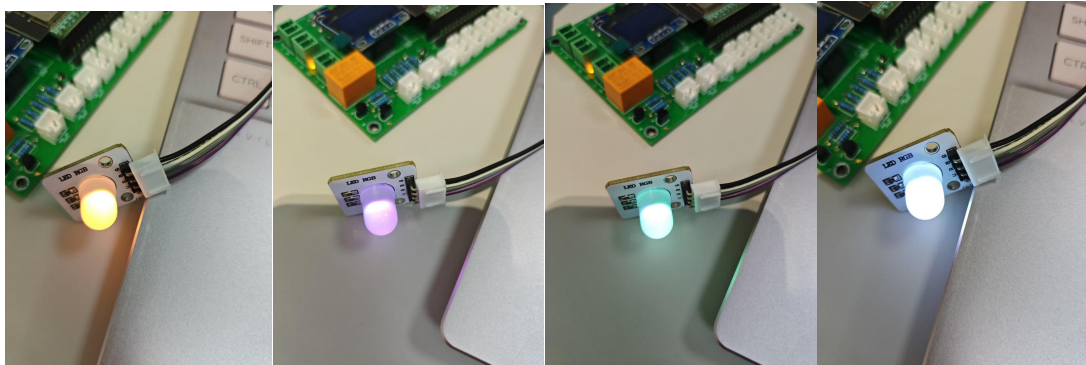
问题：因为红色、蓝色灯覆盖性强，不能产生标准黄色、紫色、青色。多次调整 RGB 比例后，可以产生明显黄色、紫色、青色。代码如下：

```
void loop() {
```

```

    analogWrite(RED, 200);
    analogWrite(GREEN, 250);
    analogWrite(BLUE, 0);
    delay(1000);
    analogWrite(RED, 230);
    analogWrite(GREEN, 0);
    analogWrite(BLUE, 200);
    delay(1000);
    analogWrite(RED, 0);
    analogWrite(GREEN, 200);
    analogWrite(BLUE, 200);
    delay(1000);
    analogWrite(RED, 255);
    analogWrite(GREEN, 255);
    analogWrite(BLUE, 255);
    delay(1000);
    analogWrite(RED, 0);
    analogWrite(GREEN, 0);
    analogWrite(BLUE, 0);
    delay(1000);
}

```



2. 产生颜色从深黄色到浅紫色（近乎白色）的呼吸灯

```

void loop() {
    analogWrite(RED, 200);
    analogWrite(GREEN, 200);
    for (int fadeValue = 0; fadeValue <= 255; fadeValue += 50) {
        analogWrite(BLUE, fadeValue);
        delay(300);
    }
    delay(1000);
}

```

成功实现呼吸灯颜色从深黄色到浅紫色（近乎白色）均匀、周期性变化。

